■ Part

expr[[*i*]] gives the i^{th} part of *expr*.

expr[[-*i*]] counts from the end.

expr[[0]] gives the head of *expr*.

expr[[*i*, *j*, ...]] is equivalent to *expr*[[*i*]] [[*j*]] ....

expr[[{i_1, i_2, ...}]] gives a combination of the parts i_1, i_2, . . . of *expr*.

You can make an assignment like *t*[[*i*]] = *value* to modify part of an expression. ■ When *expr* is a list, *expr*[[{i_1, i_2, ...}]] gives a list of parts. In general, the head of *f* is applied to the list of parts. ■ You can get a nested listing of parts from *expr*[[$list_1$, $list_2$, ...]]. Each part has one index from each list. ■ Notice that lists are used differently in Part than in MapAt and Position. ■ *expr*[[Range[*i*, *j*]]] can be used to extract sequences of parts. ■ See page 178. ■ See also: First, Head, Last, MapAt, Take.

Web sample page from The *Mathematica* Book, Second Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-51502-4; softcover ISBN 0-201-51507-5). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.